

LG Electronics: Establishing a hybrid cloud- based AI quality inspection platform at manufacturing sites with Anthos

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About LG Electronics

LG Electronics is South Korea's leading multinational electronics company that leads technological innovation in the fields of electronics, home appliances, and automobile parts around the world. The company also produces TVs, washing machines, refrigerators, auto parts, signage, robots, among others, in more than 140 countries. LG Electronics is attracting attention from the global market not only as a leading company that advances and goes beyond technological innovation but also as a global leader that continues to thrive despite turbulent times.

LG Electronics Pro Engineering Research Institute developed Cloud, an AI quality inspection platform to accelerate digital

transformation in manufacturing since the world and improve the quality of LG Electronics products.

Industries: Manufacturing

Location: South Korea



Google Cloud results

- Enables a 95% reduction in defect detection errors on the production floor
- Rapidly and reliably deploys updated applications to more production environments
- Establishment of container deployment environment and Kubernetes

Access to specialized quality inspection globally manufacturing lines MAV

About CloudMate

CloudMate is a technology-centered cloud specialist, with specialized cloud engineers and developers in each field, and helps customers use the cloud well. Rather than simply talking about cloud services and functions, we provide hands-on experience and comprehensive services, including consulting tailored to the customer's business. In addition, we are a partner with proven consulting and technical capabilities who were the first to provide Google Cloud's Anthos to Korean customers.

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cluster
management
environment
enables
integrated
management
in multiple
sites

- Completion of
an AI quality
inspection
platform
applicable to
various hybrid
cloud
environments
- Expands
business
opportunities
with the
construction
of a cloud-
based AI
quality
inspection
platform

LG Electronics (<https://www.lg.com/global/>) is leading the global market by introducing premium technologies to the smart home appliance market, including but not limited to OLED TVs, refrigerators, and washing machines. In order to apply outstanding technologies to its products and deliver products with innovative value to global customers,

it is important to build an optimal production environment that reflects stability and efficiency, while producing the latest developments in technology.

LG Electronics has been making great efforts at the manufacturing site to improve the quality of every component of its products, to ensure that it delivers finished products of the highest quality. The Manufacturing Artificial Intelligence Solution team of LG Electronics Production Engineering Research Institute is leading the digital transformation of manufacturing sites by developing MAVIN-Cloud, an artificial intelligence quality inspection platform, with Anthos and applying AI quality inspection processes through the platform to LG Electronics' manufacturing lines around the world.

Since the fourth industrial revolution, AI-related technologies have been rapidly applied to manufacturing sites. Deep learning-based machine vision inspection technology, in particular, which uses images to identify and detect defects, is increasingly being utilized in the quality inspection stages, where the surfaces and the assembly status of all the components or finished products undergo thorough examination. This is because there are many variables at the production site, and there are defects that cannot be determined only by rule-based machine vision inspection without AI. For example, defects with scratches on one part of a product or unspecified dents on the surface of a product have different locations and shapes, so standardized rule-based machine vision inspection may not be able to completely identify these types

of defects. As such, these defects would still have to be visually inspected by humans. However, when humans are involved, human errors are inevitable for various reasons, such as physical fatigue, different levels of competence in dealing with tasks, carelessness, and a lack of knowledge, among others. Deep learning technology has become the key to solving this problem.

Establishment of a machine learning-based quality inspection platform that can be quickly applied to various manufacturing lines

As machine learning-based AI technology matures, more types of defects can be accurately detected by AI on the production floor. It has become possible to determine the types of defects that were not easily identified before, with improved accuracy to skillfully catch defects that even humans would miss, as they may not be visible to the human eye. LG Electronics Production Engineering Research Institute decided to actively utilize these AI-based machine vision technologies. It plans to create an AI-based inspection model through deep learning with various examples and data of defects, and based on this, judge various types of defects that were only possible to identify with the human eye.

"LG Electronics produces a wide variety of products and has many production plants not only in Korea but also abroad. In addition, quality control is extensively managed not only in the finished product, but also in the areas where each part is

made before it becomes a finished product. AI-based quality inspection is a way to secure consistency in quality inspection in these various places and directly address the need for automation," says Hyejin Lee, Head of AI Platform Part of Manufacturing AI Solution Team at LG Electronics Production Engineering Research Institute.

Lee said that LG Electronics Production Engineering Research Institute has started machine vision-based AI quality inspection with the expectation that production sites will experience unprecedented changes. In addition, MAVIN-Cloud, LG Electronics' AI-based quality inspection platform, can cover the entire manufacturing operation environment and manage the full development and operation lifecycle of inspection models. It can create an AI inspection model suitable for each process as well.

Data and AI experts at LG Electronics used machine learning to create an AI inspection model suitable for the quality inspection process of each product and component. However, since AI inspection model development takes place in different development environments, it was not easy to distribute the inspection model to various sites at once or individually conduct retraining for maintenance afterwards. For smoother operations, each person had to configure their own development environment for developing their own deep learning model. Many AI experts were needed as well because they knew how to acquire and preprocess data and to train and evaluate models. They were

the ones who could proficiently utilize artificial intelligence technology.

The Manufacturing AI Solution Team at the LG Electronics Production Engineering Research Institute began to discuss and address a particular concern and highlighted the need to simplify the processes of training and operating machine learning models, so that anyone can use AI in many different fields, even if they are not data/AI experts. Since the training methods of machine learning models required in production sites have similar characteristics, the team decided to develop its application so that even non-experts could easily use it to train their own machine learning models. "We have developed an application for those who have a large amount of images obtained at the manufacturing sites, but are not necessarily AI experts. This application allows them to easily classify defective images or indicate that there is a defect in a specific part of a product. With just a few clicks on the mouse and with no knowledge of coding or any programming language, anybody can now train a machine learning model easily and fast through our application," says Lee.

Lee said that the way that training and managing numerous artificial intelligence quality inspection models suitable for various sites were individually conducted made it difficult to quickly apply machine learning models to various processes and many sites and properly operate them. She said what LG Electronics needed the most was a cloud-based AI quality inspection platform because with the platform's own features, the AI quality inspection

models could be efficiently and continually applied to LG's manufacturing lines and therefore, appropriately well-maintained.

"The ultimate goal was to create a common, standardized AI quality inspection platform across the enterprise that enables even people without expertise in AI to train and use deep learning models directly only with very little training and to share the models and related data with each other."

Spreading the platform to more environments from the start with cloud

The Manufacturing AI Solution Team at the LG Electronics Production Engineering Research Institute is an organization that develops various AI-based manufacturing solutions and platforms to improve the manufacturing competitiveness of LG Electronics and the entire LG Group.

As such, from the very moment the team began to think about and discuss the structure of MAVIN-Cloud, the team made it a priority that all of its platform users, regardless of whether they are from LG Electronics or not, could have the same experience.

Most actual data generated and obtained at manufacturing sites are a valuable resource that cannot be shared outside the enterprise. For this reason, each site had to be able to create an AI quality inspection model to be used in the production line through the individually built MAVIN-

Cloud without exporting the data stored in the on-premise environment constructed independently for each site. However, in the case of building individual MAVIN-Cloud services in different on-premise environments for each site, a significant amount of maintenance and management was required and the job was never easy because updating and distributing applications and verifying the work had to be continuously performed onsite.

After numerous discussions internally, a consensus was reached to prepare to provide the same service to each site through the configuration of a Kubernetes cluster. However, because the security, network, and management system of the on-premise environment at each site were different, there was a remaining assignment to maintain the Kubernetes cluster environment, and Anthos (<https://cloud.google.com/anthos/>) by Google Cloud (<https://cloud.google.com/>) was the service that could solve these concerns most effectively.

Lee said that because Anthos makes it possible to equally manage all Kubernetes clusters in multiple environments, it ensures consistency in management, such as application deployment and operation.

"Through Anthos, we deployed MAVIN-Cloud equipped with improved functions with a new algorithm, laying the groundwork for the same operation in all environments."

Stable service distribution and operation management,

introduced to more manufacturing sites, a 95% reduction in defect detection errors

In order for MAVIN-Cloud to fully fulfill its role as an AI-based quality inspection platform, fast and safe deployment, continuous updates and operational management had to be implemented practically. The Manufacturing AI solution team decided that MAVIN-Cloud, to be operated at LG Electronics, should be constructed in the way that an on-premise Kubernetes cluster, on which applications and services are deployed, is built, and the platform provides a Software-as-a-Service-like environment for the end users.

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The Manufacturing AI Solution Team first built an HPC-level, high-performance, on-premise-based private cloud environment, and prepared an environment to operate MAVIN-Cloud based on Anthos. The team placed their trust in Google Cloud for its technology, and in the way it handles Kubernetes containers. Lee said that Anthos connects the on-premise and cloud environments, which enables consistency in deployment and management as well as enhances the stability of MAVIN-Cloud by providing continuous technical support at the manufacturing site.

"Anthos runs on Google Kubernetes Engine. Because Kubernetes is designed by Google and operated in an open-source format, new elements of Kubernetes are applied quickly, and optimization is achieved well. Container technology is still developing rapidly, and stability is just as important as new technology at production and manufacturing sites. Anthos was able to quickly reflect the latest elements while securing high stability based on the stability of Google Cloud and Kubernetes technology," says Lee.

"Through
MAVIN-Cloud,
the development
period of the AI
quality
inspection
model has been
shortened by
more than 50%,
and the rate of
false defect
judgments in the
production line
to which the AI
quality
inspection

model is applied
has been
reduced by 95%.
Accuracy of
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MAVIN-Cloud is stable and is continuously delivering the value of an AI quality inspection platform to users through continuous advanced development and regular distribution. LG Electronics used MAVIN-Cloud as a standard quality inspection platform for the company's manufacturing sites around the world, and the effects were immediately noticeable. While human error can occur in the process of checking defects in parts or products, machine learning has helped to ensure a higher level of

quality control standards. Lee explains that the accuracy of the quality inspection model is rapidly improving due to the effect of continuous AI algorithm advancement and algorithm retraining.

"Through MAVIN-Cloud, the development period of the AI quality inspection model has been shortened by more than 50%, and the rate of false defect judgments in the production line to which the AI quality inspection model is applied has been reduced by 95%. Accuracy of Defect Decisions has been increasing, and the loss caused by wrong judgment has been greatly reduced. Above all, Anthos has provided a stable foundation for service distribution and operation to further expand the scope of application of AI-based quality inspection, thereby helping LG Electronics rapidly accelerate and apply AI quality inspection to its production lines around the world," says Lee.

Flexibility to apply to any site, expand new opportunities with public services

The greatest strength of Anthos is its flexibility to connect to any cloud. Once the Anthos framework, which creates a Google Cloud-like environment, is installed on the system, it can be operated on any hardware. This allows organizations to utilize their existing servers, or their surplus resources. In addition to on-premise, it is flexible enough to connect with the existing public cloud.

"Anthos has created an environment that can continuously develop, distribute, and efficiently manage MAVIN-Cloud, an AI quality inspection platform, that is widely used at LG Electronics' production sites not only in Korea but also around the world.

Anthos helped us quickly and reliably deploy the latest technologies and applications to meet the needs of

manufacturing
sites based on
the stability of
Google Cloud
and technology
in Kubernetes.
LG Electronics
Production
Engineering
Research
Institute goes
beyond
developing and
operating an
artificial
intelligence
quality
inspection
platform utilized
at the company's
production sites
and is preparing
to expand the
business by
launching a
public cloud-
based MAVIN-
Cloud service

that will enable
the introduction
and operation of
AI quality
inspection
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industrial sites
outside of LG
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*—Hyejin Lee, Head of AI
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This characteristic of Anthos has become a new opportunity for LG Electronics Production Engineering Research Institute. MAVIN-Cloud is being rapidly applied to LG Electronics' production lines around the world. Expansion into other fields is also underway. Preparations are currently being made to apply MAVIN-Cloud to other affiliates within the LG Group, and services that can be used universally in various industrial sites besides LG Group companies are also being reviewed.

To this end, we have configured it to be used not only on-premises but also with public clouds, including Compute Engine, for machine learning and field operations. Because Anthos has the flexibility to embrace a variety of field environments, MAVIN-Cloud can respond to quality inspections at any industrial site as long as appropriate computing resources are secured.

The collaboration with partner CloudMate (<https://cloudmt.co.kr/>) also played an important role in enabling the manufacturing AI solution team to successfully apply MAVIN-Cloud to production lines in various environments and secure flexibility. The Manufacturing AI Solution Team, along with the partner, addressed their concerns through a proof of concept based on various scenarios for deployment and field operation from the Kubernetes introduction stage, and as a result, the team was able to smoothly introduce it into the field.

In addition, the partner performs tasks, such as responding to and monitoring infrastructure failures and managing system resources, while the Manufacturing AI Solution Team focuses on platform development and distribution, such as algorithm updates and functional enhancements. The team has established a collaborative environment where MAVIN-Cloud can be stably introduced to more sites and operated well and effectively, and therefore, the team has been able to continuously provide services that users are satisfied with.

"Because the AI quality inspection platform is successfully operated at LG Electronics through Anthos, the team has not only enhanced the vision inspection capabilities but also built the vibration and noise inspection capabilities, thereby expanding the business, and the team is planning to provide services in the form of a SaaS-type cloud platform outside of LG Electronics," says Lee.

Lee said that through Anthos, she was able to build the cloud-based AI quality inspection platform she had originally envisioned, and gained confidence through successful deployment and operation, adding that with the help of Anthos and Google Cloud, the LG Electronics Production Engineering Research Institute will expand the business that goes beyond LG Electronics. The Manufacturing AI Solution Team will soon be launching a public cloud-based MAVIN-Cloud service that will enable the introduction and operation of AI quality inspection processes in many more industrial sites around the world. The team's experience in field application of the hybrid cloud has led to expansion into new businesses.